

N-Party Cyclic Atomic Swap

Overview and Assumptions

For stakeholders

April 13, 2026

1 Protocol Overview

This protocol allows N parties to simultaneously exchange assets across different blockchains, without any trusted intermediary. The parties form a directed cycle: each party deposits funds that the next party in the cycle can claim, and in turn claims the funds deposited by the previous party.

Example. Three parties A , B , C want to trade: A sends Bitcoin to B , B sends Cardano ADA to C , C sends Bitcoin to A . The protocol guarantees that under certain assumptions (discussed below) either *all three transfers happen* or *none of them do*: no party can lose funds unless they violate their own obligations.

How It Works

1. **Agree.** All parties agree on the swap terms: who trades what with whom, and the safety timing parameters (refund windows). One party is designated the **leader**.
2. **Prepare.** Through a series of message exchanges, the parties jointly construct three transactions per transfer: a *deposit* (locks funds on-chain), a *withdraw* (lets the recipient claim those funds), and a *refund* (lets the depositor reclaim if the swap fails). The withdraw transaction is incomplete at this stage: it requires a shared secret that no single party possesses.
3. **Lock.** Each party broadcasts their deposit transaction. Funds are now locked on-chain and can only move via the pre-arranged withdraw or refund transactions.
4. **Claim.** The leader collects secret shares from all other parties, completes the first withdraw transaction, and broadcasts it on-chain. This broadcast publicly reveals the shared secret. All other parties extract the secret from the blockchain and complete their own withdraw transactions.
5. **Refund (if anything goes wrong).** If the leader never broadcasts the first withdraw, each party's refund transaction becomes valid after a pre-agreed block height. Every party reclaims their own deposit. Refund windows are staggered so that no party can simultaneously collect a withdraw and a refund.

The key mechanism is that the act of claiming funds on one chain *automatically reveals* the information needed for all other parties to claim on their respective chains. This is achieved through adaptor signatures, a cryptographic technique that ties the validity of a signature to the disclosure of a secret.

2 Assumptions and Their Implications

The protocol's safety guarantees hold only when certain assumptions are met. This section lists each assumption, explains why it is needed, and describes the consequences when it is violated.

2.1 Cyclic Swap Topology

Assumption. The parties must form a single directed cycle: each party deposits into exactly one transfer and claims from exactly one transfer.

Why. The staggered refund-window mechanism that prevents double-spending (collecting both a withdraw and a refund) relies on a strict linear ordering of the parties. In non-cyclic topologies (e.g. a party that deposits into two transfers), this mechanism wouldn't work.

Implication. The protocol cannot express arbitrary multi-party trades.

2.2 Timely Chain Access

Assumption. Every party must have **timely chain access** throughout the protocol: the ability to both *read* the current state of every relevant blockchain and *write* transactions that are included in a block, all within a bounded number of blocks.

Specifically, after the trigger transaction is broadcast, each non-leader party must be able to (1) observe the trigger at the required finality depth, (2) construct and sign their own withdraw transaction, and (3) have that transaction confirmed — all within Δ blocks (the **claim propagation time**).

Why. The protocol does not tolerate situations where a party is unable to interact with the chain in time. Both consistency (all parties see the same chain state) and availability (all parties can get transactions included) are required.

What can break timely chain access:

- **Party downtime** — node crash, key unavailability, operator error.
- **Network-level censorship** — miners or validators selectively exclude a party's transactions from blocks. On PoW chains a mining cartel can censor; on PoS chains a colluding validator set can achieve the same effect.
- **Chain congestion** — sustained high demand delays transaction inclusion beyond Δ blocks even without targeted censorship. This is economically equivalent to censorship.
- **Network partitions** — a party is unable to communicate with blockchain nodes despite being locally operational.
- **Chain reorganisation** — a confirmed transaction is reversed, resetting the party's view of finality and potentially consuming Δ blocks before re-confirmation.

Implication. A party that loses chain access after the Lock phase risks losing its deposit: another party may claim the locked funds via the withdraw path, while the affected party cannot claim its own output. In practice, parties should use monitoring infrastructure (watchtowers, redundant nodes) and Δ must be set conservatively for worst-case inclusion delays. The protocol offers no defence against a censoring majority of block producers collaborating with a malicious participant.

2.3 Sufficient Refund Window Gaps

Assumption. Adjacent refund windows must be separated by at least Δ blocks: $W_i - W_{i+1} \geq \Delta$ for all i .

Why. If the gap is too small, a malicious leader can time the trigger broadcast so that a party's refund window opens before it has had enough time to confirm its withdraw transaction. The victim refunds their own deposit (believing the swap failed), but the leader also claims the victim's locked funds via the withdraw path. The victim loses their deposit and receives nothing. Any of the threats to timely chain access listed in the previous section can cause the effective propagation time to exceed Δ .

Implication. The total lock-up time grows linearly with N : the leader's refund window must be at least $(N-1) \times \Delta$ blocks after the earliest window. Choosing Δ is a trade-off between safety and capital efficiency; **underestimating Δ is a safety-critical error**. Δ should be computed per chain-pair (using the worst case) and agreed upon by all parties in the Config phase.

2.4 Chain Finality

Assumption. The protocol treats “transaction confirmed” as a binary oracle. The definition of finality (e.g. number of confirmations) is chain-specific and external to the protocol.

Implication. A chain reorganisation that reverses a confirmed deposit after the claim phase has begun can break atomicity irrecoverably. Cross-chain swaps are only as safe as the weakest finality guarantee among the chains involved.

2.5 Schnorr-Compatible Signatures

Assumption. All participating blockchains must support Schnorr-compatible signature verification (or an equivalent via smart contracts).

Why. The protocol relies on adaptor signatures, which require the algebraic structure of Schnorr signatures (specifically, the linearity of nonce randomisation). ECDSA-based adaptor signatures exist but have different security properties and are less mature.

Implication. Bitcoin supports Schnorr natively since Taproot (November 2021). Cardano uses EdDSA natively, which is not directly compatible; a smart-contract bridge is required to verify Schnorr signatures on Cardano. Chains without Schnorr support or smart-contract capability cannot participate.

2.6 Cryptographic Soundness

Assumption. The adaptor signature scheme, the MuSig2 multi-signature scheme, and their composition are assumed correct and secure.

Why. If the underlying cryptography is broken (e.g. an adversary can forge adaptor signatures or extract the aggregate secret without the trigger), the protocol’s guarantees are void.

Implication. MuSig2 has a published security proof, but the composition of MuSig2 with adaptor signatures in the specific way this protocol uses them has not been independently verified in a peer-reviewed setting. This is a residual cryptographic risk.

2.7 Mutual Exclusion of Withdraw and Refund

Assumption. For each transfer, at most one of the withdraw and refund transactions can be confirmed. Once one succeeds, the other is permanently invalid.

Why. Without this guarantee, a party could collect both: claim the withdraw (receiving the counterparty’s funds) and then also claim the refund (recovering their own deposit), stealing funds outright.

Implication. On UTXO-based chains (Bitcoin, Cardano) this holds by construction: spending a UTXO consumes it. On account-based chains (e.g. Ethereum), a smart contract must explicitly enforce this exclusion. Any chain integration must be audited for this property.

2.8 Griefing and Optionality

Assumption. The protocol guarantees that no honest party loses principal, but it does not prevent an adversary from forcing all parties to wait out their refund windows, nor does it prevent the leader from exploiting the temporal gap between commitment and completion.

Why — griefing. Any party can abort after the Lock phase by simply withholding their adaptor secret. All parties eventually recover their deposits via refund, but their funds are locked until each party’s refund window opens.

Why — optionality. Between the Lock phase (all funds committed on-chain) and the leader’s decision to broadcast the trigger, time passes and asset prices move. The leader can

observe these movements and choose to complete the swap (if prices moved in their favour) or let it expire via refund (if prices moved against them). The non-leader parties are locked in and cannot exit. Han et al. [1] prove that this is structurally equivalent to a premium-free American call option granted to the leader at the non-leaders' expense, and estimate the fair option premium at 2–3% of the swap value for typical cryptocurrency volatility. This is a structural property of all atomic swap protocols in which one party commits last.

Implication. Griefing imposes asymmetric costs: a party with an early refund window can force the leader to wait at low cost to themselves. Optionality transfers expected value from non-leaders to the leader, since the leader can choose to complete or abort based on price movements. The protocol has no built-in penalty for griefing and no premium mechanism for optionality; mitigation (reputation systems, bonds, trigger deadlines, option premiums) is outside the protocol scope.

2.9 No Upper Bound on Transaction Validity

Assumption. Bitcoin (and most blockchains) only support a *lower* bound on when a transaction becomes valid (“not before block n ”), not an upper bound (“not after block m ”).

Why. This means that once a withdraw transaction is valid, it remains valid *forever*. There is no way to make it expire.

Implication. Once the refund window opens, both withdraw and refund are simultaneously valid (mutual exclusion ensures only one succeeds). A party must broadcast their refund promptly; delay risks the counterparty completing the withdraw first. This reinforces the timely chain access requirement (Section 2.2).

2.10 Pre-Determined Leader

Assumption. One party is designated as the leader before the protocol begins. The leader triggers the claim phase by broadcasting the first withdraw transaction.

Why. The refund windows are anchored to the leader's position: the leader's refund window opens last, giving all other parties the maximum time to observe the trigger and claim. Changing the leader mid-protocol would invalidate the window ordering.

Implication. Leader selection must happen out-of-band or via a pre-protocol election mechanism (not specified). The leader role carries both costs and benefits:

- **Cost:** The leader's funds are locked for the longest duration, and the leader bears the highest opportunity cost if the swap is aborted (see Section 2.8).
- **Benefit:** The leader holds the free option described in Section 2.8: they can observe price movements after all funds are locked and choose whether to complete or abort. This optionality transfers expected value from non-leaders to the leader.

These two effects pull in opposite directions: griefing risk discourages taking the leader role, while optionality rewards it. Whether the net incentive is positive or negative depends on asset volatility, lock-up duration, and the number of parties. The protocol does not balance these incentives; rational leader selection is an open design problem.

References

- [1] Runchao Han, Haoyu Lin, Jiangshan Huang, Zhimei Zhang, and Hong Zhu. On the optionality and fairness of atomic swaps. In *Proceedings of the First ACM Conference on Advances in Financial Technologies (AFT '19)*. ACM, 2019. Also available as IACR ePrint 2019/896.